AI-Powered Geoacoustic Fingerprinting for Conflict Mineral Traceability in South Sudan

BSc. in Software Engineering

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Date: 28th, Jan, 2026

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LIST OF ACRONYMS / ABBREVIATIONS

Acronym	Meaning
AI	Artificial Intelligence
ML	Machine Learning
CNN	Convolutional Neural Network
DSR	Design Science Research
RFID	Radio Frequency Identification
GPS	Global Positioning System
XRF	X-ray Fluorescence
REE	Rare Earth Elements
ICGLR	International Conference on the Great Lakes Region
UNEP	United Nations Environment Programme
API	Application Programming Interface
PWA	Progressive Web Application

JSON	JavaScript Object Notation
MFCC	Mel-Frequency Cepstral Coefficients
UML	Unified Modeling Language
SRS	Software Requirements Specification
FR	Functional Requirements
NFR	Non-Functional Requirements

CHAPTER ONE: INTRODUCTION

1.1 Background of the Study

Mineral resources play a central role in the global economy, supporting critical sectors such as construction, electronics, renewable energy technologies, and advanced manufacturing. Demand for minerals has continued to rise alongside global industrialization and the transition toward low-carbon and digital economies, **Ferrin, T. E. (2020).** For developing and post-conflict countries, mineral extraction is often viewed as a pathway for economic growth, employment creation, and national revenue generation **Vijeyarasa, R. (2020, July 7)**.

At the international level, growing awareness of ethical sourcing and responsible production has led to increased scrutiny of mineral supply chains. Global and regional initiatives now emphasize transparency, due diligence, and accountability to ensure that minerals are sourced responsibly and in compliance with human rights and environmental standards. **Buhmann, K. (2021, September 20)** These frameworks reflect a broader shift toward conflict-sensitive and sustainability-driven resource governance, particularly for minerals originating from fragile and conflict-affected regions.

To support these governance objectives, technological solutions have increasingly been adopted within mineral supply chains. Early traceability approaches relied heavily on manual inspections, paper-based records, and periodic audits. Over time, these methods were supplemented by digital databases, logistics tracking systems, and centralized reporting platforms. More recently, blockchain-based traceability systems have been introduced to enhance data transparency, immutability, and auditability across complex, multi-stakeholder supply chains. **Mubarik, M. (2022, March 22)**. Empirical studies highlight that while such systems strengthen record management and coordination, they primarily focus on documenting transactions rather than analyzing the physical characteristics of minerals themselves. **Kirchherr, J. (2019, September 26)**

In parallel with supply chain digitization, advances in artificial intelligence (AI) and machine learning (ML) have transformed analytical practices in geology, materials science, and mineral exploration. Recent studies demonstrate that machine learning models can identify complex patterns within mineralogical, chemical, and visual datasets, enabling automated mineral classification and compositional analysis with high accuracy. **Liu, H. (2021, November 30)**. These developments indicate strong potential for AI-driven techniques to support mineral verification and characterization tasks.

Geological research further demonstrates that minerals formed under different environmental and geological conditions exhibit measurable variations in trace element composition, microstructure, and physical properties. Such variations have been successfully used as provenance indicators in controlled analytical settings. **Box, L. (2020, April 18)**. Alongside geochemical methods, acoustic emission and resonance-based studies show that solid materials produce characteristic acoustic responses influenced by internal structure and composition, which can be captured and analyzed using data-driven techniques. **Anoyatis, G. (2023, May 8)**

Recent progress in multi-modal machine learning has enabled the integration of heterogeneous data sources, including images, numerical measurements, and signal-based features. Studies show that multi-modal fusion approaches improve robustness, generalization performance, and resilience to noisy or incomplete data when compared to single-modality models. **Kauttonen, J. (2024, April 6)**. These advances open new possibilities for intrinsic, data-driven approaches to mineral verification that go beyond document-based traceability mechanisms.

Building on these technological and scientific developments, this study explores how an AI-powered, multi-modal geoacoustic fingerprinting approach can contribute to mineral provenance verification in conflict-affected and low-resource environments, with a specific focus on South Sudan.

1.2 Problem Statement

Despite the adoption of digital traceability technologies, South Sudan lacks a reliable, scalable, and scientifically grounded method for verifying the geological origin of minerals. Existing traceability systems primarily track documentation, transactions, and transportation routes rather than examining the intrinsic physical and compositional properties of the minerals themselves. As a result, minerals originating from illegal or conflict-linked mining sites can be falsely registered as legitimate and introduced into global supply chains without detection **Manamela, D. (2019)**

Blockchain-based traceability platforms and GPS-enabled logistics systems have improved transparency and record integrity but remain entirely dependent on accurate data entry at the point of origin. Recent studies show that when fraudulent or incorrect information is recorded during initial registration, these systems are unable to detect deception and instead propagate inaccurate records throughout the supply chain. **Luty, P. R. (2022)**.

Hardware-dependent verification approaches, such as RFID tagging and laboratory-based spectroscopic analysis, face additional barriers in artisanal mining contexts, including high costs, limited infrastructure, power constraints, and susceptibility to tampering. **Bester, V., & Uys, T. (2023, June 6)**. Consequently, there is currently no widely deployed system capable of providing objective, scientific verification of mineral provenance that is feasible in low-infrastructure and conflict-affected environments.

This gap enables the laundering of conflict minerals, undermines peace-building efforts, and excludes legitimate artisanal miners from responsible international markets. Addressing this challenge requires a shift from document- and logistics-based traceability toward verification approaches grounded in the intrinsic physical, chemical, and structural characteristics of minerals.

1.3 Aim of the Study

This study aims to design and evaluate an AI-powered geoacoustic fingerprinting system that enables scientific verification of mineral provenance in South Sudan through multi-modal machine learning.

1.4 Objectives of the Study (SMART)

Main Objective

To develop and evaluate a software-based, multi-modal AI system capable of generating unique mineral fingerprints for provenance verification in conflict-affected mining environments.

Specific Objectives

1. To conduct a systematic review of peer-reviewed literature on conflict mineral traceability, geological provenance, acoustic analysis, and multi-modal machine learning within the first three weeks of the project.

2. To design and implement a multi-modal dataset integrating mineral images, chemical composition data, and acoustic features using secondary and simulated sources within a three-month project timeline.

3. To develop a machine learning model that generates a persistent and unique fingerprint for each registered mineral sample.

4. To evaluate the system using measurable performance metrics, including accuracy, precision, recall, and false-positive rate.

1.5 Research Questions

1. Can multi-modal machine learning effectively distinguish minerals originating from different geological locations with sufficient accuracy to support provenance verification?

2. Which combination of visual, chemical, and acoustic features provides the highest discriminative power for mineral fingerprinting?

3. How does the proposed AI-based verification approach compare with existing document-based traceability systems in terms of accuracy and feasibility in low-resource environments?

1.6 Scope of the Study

- **Geographical Scope:** Selected mining regions in South Sudan, including Kapoeta East, Central Equatoria, and the Yei River area.

- **Content Scope:** Gold, copper-bearing minerals, and iron ore are commonly associated with artisanal mining.

- **Technical Scope:** A software-based prototype using secondary and simulated datasets without reliance on specialized field hardware.

- **Time Scope:** January 2026 – March 2026.

1.7 Significance of the Study

This research contributes to conflict mineral governance by proposing a scientific, data-driven approach to mineral provenance verification that complements existing regulatory and traceability frameworks. Practically, the system has the potential to support legitimate artisanal miners by enabling access to responsible and ethical supply chains. Academically, the study advances software engineering and applied machine learning research by demonstrating how multi-modal AI systems can be designed for deployment in low-resource and conflict-affected environments.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

This chapter reviews recent peer-reviewed literature related to conflict mineral governance, digital traceability systems, geological provenance techniques, machine learning-based mineral analysis, acoustic material characterization, and multi-modal artificial intelligence. The purpose of this review is to situate the proposed research within current academic discourse, critically evaluate existing approaches, and identify unresolved gaps that motivate the development of an AI-powered geoacoustic fingerprinting system.

The review draws exclusively from **peer-reviewed journals, conference proceedings, and academic books published between 2019 and 2026**, reflecting the most current advances in mineral traceability, machine learning, and computational geoscience. The literature is organized thematically to reflect both technological evolution and methodological relevance to conflict-affected, low-infrastructure environments such as South Sudan.

2.2 Conflict Minerals and Contemporary Governance Challenges

Recent studies continue to highlight the persistent link between natural resource extraction, weak governance, and armed conflict in fragile states. **Wagner, N. (2025)** emphasizes that artisanal and small-scale mining remains a critical livelihood activity in post-conflict regions but is often poorly regulated, creating opportunities for mineral laundering and exploitation.

More recent governance-focused research has shifted from descriptive analyses of conflict minerals toward evaluating the effectiveness of regulatory frameworks. According to **Leuz, C. (2021),** international certification schemes and due diligence requirements have improved corporate awareness but have not eliminated illicit mineral flows, particularly in regions with limited institutional capacity.

Studies focusing on East and Central Africa further show that governance-based traceability systems rely heavily on manual reporting, self-declaration, and periodic audits, which are difficult to enforce in remote mining areas. **Dept, I. M. F. A. (2022)**. These findings indicate that governance mechanisms alone are insufficient to guarantee the physical authenticity of mineral origin, especially in artisanal mining contexts.

2.3 Digital Traceability Systems: Blockchain, IoT, and Logistics Technologies

Blockchain technology has received significant attention in recent years as a solution for improving transparency in mineral supply chains. **Venkatesh, M. (2021)** and **Rossignoli, C. (2021, March 30)** demonstrate that blockchain can enhance auditability, reduce record tampering, and improve stakeholder coordination. However, empirical evaluations show that blockchain systems do not verify the correctness of the data entered at the source.

More recent work by **Casino, F. (2022)** emphasizes that blockchain-based traceability suffers from the "data authenticity problem," whereby the system cannot detect fraudulent information entered during initial registration. This limitation is particularly problematic in conflict-affected regions where trust in institutions is low.

IoT-based solutions, including RFID and GPS tracking, have also been deployed to monitor mineral transportation. Research by **Celik, T. (2024, May 7)** indicates that while such systems improve logistical visibility, they remain vulnerable to tag removal, substitution attacks, and equipment failure. Importantly, these technologies track containers rather than verifying the mineral itself, leaving a critical gap in provenance verification.

2.4 Geological Provenance and Mineral Fingerprinting Techniques

Recent advances in applied geochemistry reaffirm that minerals formed in different geological environments exhibit measurable variations in trace elements, isotopic ratios, and microstructural properties. **Mir, A. R. (2023, August 3)** demonstrates that rare earth element (REE) signatures can be used to distinguish mineral sources with high confidence under controlled conditions.

More recent studies by **Zhang, Y. (2020)** show that multi-element geochemical fingerprints can improve provenance accuracy when combined with statistical modeling. However, these techniques typically require laboratory-grade equipment, expert interpretation, and controlled sampling conditions.

Despite their scientific validity, recent reviews emphasize that laboratory-based provenance methods remain impractical for large-scale deployment in artisanal and low-resource mining environments **Ambrosio, M. (2019)**. As a result, there is growing interest in software-driven approaches that approximate geological fingerprinting without dependence on advanced laboratory infrastructure.

2.5 Machine Learning Applications in Mineral and Geological Analysis

Machine learning has seen increased adoption in mineralogical and geological analysis over the past decade. Recent studies demonstrate that supervised learning and deep learning models can effectively classify minerals using chemical, spectral, and visual data. **Wolverton, C. (2022)** showed that convolutional neural networks achieve high accuracy in mineral identification using Raman spectroscopy. **Wang, X. Z. Y. X. X. J. G. (2021)** extended this work using image-based CNN architectures for automated mineral classification.

More recent work by **Abdelsamea, M. M. (2023)** highlights the potential of machine learning to detect subtle compositional patterns that are difficult to identify using traditional statistical methods. However, most existing machine learning applications focus on **mineral type classification**, not **geographical provenance**. Models are trained to answer *what mineral is this?* rather than *where did this mineral come from?*

This limitation is repeatedly acknowledged in recent computational geoscience literature, which calls for new research directions focused on location-specific fingerprinting rather than categorical identification **Punmiya, R. (2021)**

2.6 Acoustic Analysis and Material Characterization

Acoustic analysis has long been used in materials science to assess internal structure, density, and mechanical properties. Recent studies extend classical acoustic emission theory to data-driven modeling approaches. **Westerhausen, C. (2019)** demonstrate that acoustic responses contain discriminative features linked to microstructural variation.

More recent research by **Garcke, J. (2020)** shows that machine learning models trained on acoustic resonance data can distinguish between materials with similar visual or chemical properties. These findings suggest that acoustic signals can serve as complementary descriptors of material identity.

However, existing acoustic studies focus primarily on industrial materials, construction elements, or laboratory rock samples. The use of acoustic features for **mineral provenance verification**, particularly in combination with other data modalities, remains largely unexplored.

2.7 Multi-Modal Machine Learning and Data Fusion

Multi-modal machine learning has emerged as a powerful paradigm for integrating heterogeneous data sources. Surveys by **Clifton, D. A. (2023)** and more recent work by **Chen, L. (2020**) show that combining visual, numerical, and signal-based inputs improves robustness, generalization, and resistance to noise.

Recent studies in applied sciences demonstrate that multi-modal fusion is particularly effective when individual modalities are incomplete or unreliable. According to **Wei, H., & Hu, J. (2023)**, intermediate-level fusion architectures allow models to learn cross-modal correlations that are not observable in unimodal systems.

Despite these advances, there is minimal research applying multi-modal AI to **conflict mineral provenance verification**. Existing work either applies multi-modal learning in unrelated domains or focuses on classification tasks without addressing governance or traceability challenges.

2.8 Research Gap and Chapter Summary

The reviewed literature reveals a clear and well-defined research gap. Contemporary traceability systems improve documentation and logistics but do not verify the physical origin of minerals. Geological provenance techniques confirm that location-specific signatures exist but remain inaccessible in low-resource environments. Meanwhile, recent advances in machine learning, acoustic analysis, and multi-modal data fusion demonstrate strong analytical potential but have not been systematically applied to conflict mineral verification.

There is currently **no software-based, multi-modal AI system** designed to generate unique mineral fingerprints for provenance verification in artisanal and conflict-affected mining contexts. This study addresses this gap by proposing an AI-powered geoacoustic fingerprinting system that integrates visual, chemical, and acoustic features to verify mineral origin, prevent duplicate registration, and support ethical mineral supply chains in South Sudan.

CHAPTER THREE: SYSTEM ANALYSIS AND DESIGN

3.1 Introduction

This chapter presents the system analysis and design of the proposed AI-Powered Geoacoustic Fingerprinting System for Conflict Mineral Traceability in South Sudan. The purpose of this chapter is to translate the conceptual and theoretical foundations discussed in earlier chapters into a structured and implementable system design.

The chapter describes the research design, development model, system requirements, system architecture, and Unified Modeling Language (UML) diagrams that collectively explain how the proposed system will operate. Given the complexity of multi-modal machine learning systems and the constraints of low-resource and conflict-affected environments, the design emphasizes modularity, scalability, usability, and data integrity.

The study adopts a Design Science Research (DSR) methodology, focusing on the creation and evaluation of a technological artifact that addresses a real-world problem. The system design prioritizes multi-modal data integration, unique mineral identification, duplicate detection, and feasibility within an academic prototype context.

3.2 Research Design & the Development Model

3.2.1 Research Design

This study follows a Design Science Research (DSR) approach, which is appropriate for developing and evaluating innovative software systems intended to solve practical problems. DSR emphasizes problem relevance, systematic artifact creation, and rigorous evaluation.

The primary artifact in this research is a software-based system that applies artificial intelligence to generate and verify unique mineral fingerprints using multimodal data. A quantitative experimental design is used to evaluate system performance. Metrics such as accuracy, precision, recall, and false-positive rate are employed to assess the effectiveness of the fingerprinting and verification process.

3.2.2 Development Model

The system development follows a hybrid Agile–Waterfall model, combining structured planning with iterative experimentation. This approach is well-suited for machine learning projects where system requirements are defined early, while data preprocessing and model optimization require flexibility.

The development process is structured as follows:

Requirements and Planning Phase (Waterfall):
System scope, functional and non-functional requirements, data modalities, and evaluation criteria are clearly defined.

Iterative Development Phase (Agile):
Data preprocessing pipelines, feature extraction modules, machine learning models, and verification logic are developed incrementally through short iterations.

Evaluation and Documentation Phase (Waterfall):
Final testing, performance evaluation, and academic documentation are completed to ensure clarity, reproducibility, and alignment with research objectives.

3.2.3 Proposed Model (Mission-Customized Description)

The proposed system model consists of the following stages:

- Registration of certified mining sites and reference minerals
- Mineral scanning and metadata capture
- Multi-modal feature extraction (visual, chemical, and acoustic)
- Feature fusion and geoacoustic fingerprint generation
- Unique mineral identity assignment and storage
- Provenance verification and duplicate detection
- Result reporting and audit trail generation

This model shifts mineral traceability from document-based tracking to intrinsic, data-driven verification.

3.3 Software Requirements Specification(SRS)

This section defines the functional and non-functional requirements of the system. These requirements guide the system design and ensure alignment with the research objectives and proposed solution.

3.3.1 Functional Requirements

Req ID	Requirement	Description
FR-01	User Authentication	The system shall allow authorized users (inspectors, regulators, auditors) to log in using role-based access control securely
FR-02	Mining site registration	The system shall allow administrators to register certified mining sites with location and certification details.
FR-03	Mineral registration	The system shall allow minerals to be registered and linked to a certified mining site before verification.

FR-04	Mineral scanning	The system shall allow users to scan minerals using a mobile or web device to capture images and scan metadata.
FR-05	Multi-modal data processing	The system shall process mineral images, acoustic signals, and chemical composition data as inputs to the verification pipeline.
FR-06	Feature extraction	The system shall extract visual, acoustic, and chemical features from raw input data.
FR-07	Fingerprint generation	The system shall generate a unique geoacoustic fingerprint for each registered mineral sample.
FR-08	Duplicate detection	The system shall detect repeated scans of the same mineral using scan hashes and fingerprint similarity.
FR-09	Provenance verification	The system shall compare generated fingerprints with stored reference fingerprints to verify mineral origin.
FR-10	Verification result reporting	The system shall display verification results and confidence scores to the user.
FR-11	Verification record storage	The system shall store verification outcomes, timestamps, and user information for audit purposes.
FR-12	Audit trail management	The system shall maintain a complete and immutable history of scans and verifications

The functional requirements describe the system's core capabilities, including user authentication, mineral and mining site registration, multi-modal data processing, fingerprint generation, duplicate detection, provenance verification, and audit trail management.

3.3.2 Non-Functional Requirements

Requirement Type	Req ID	Description
Security	NFR-01	The system shall ensure secure access through authentication, authorization, and encrypted communication.
Usability	NFR-02	The system shall provide a simple and intuitive user interface suitable for non-technical users in field environments.

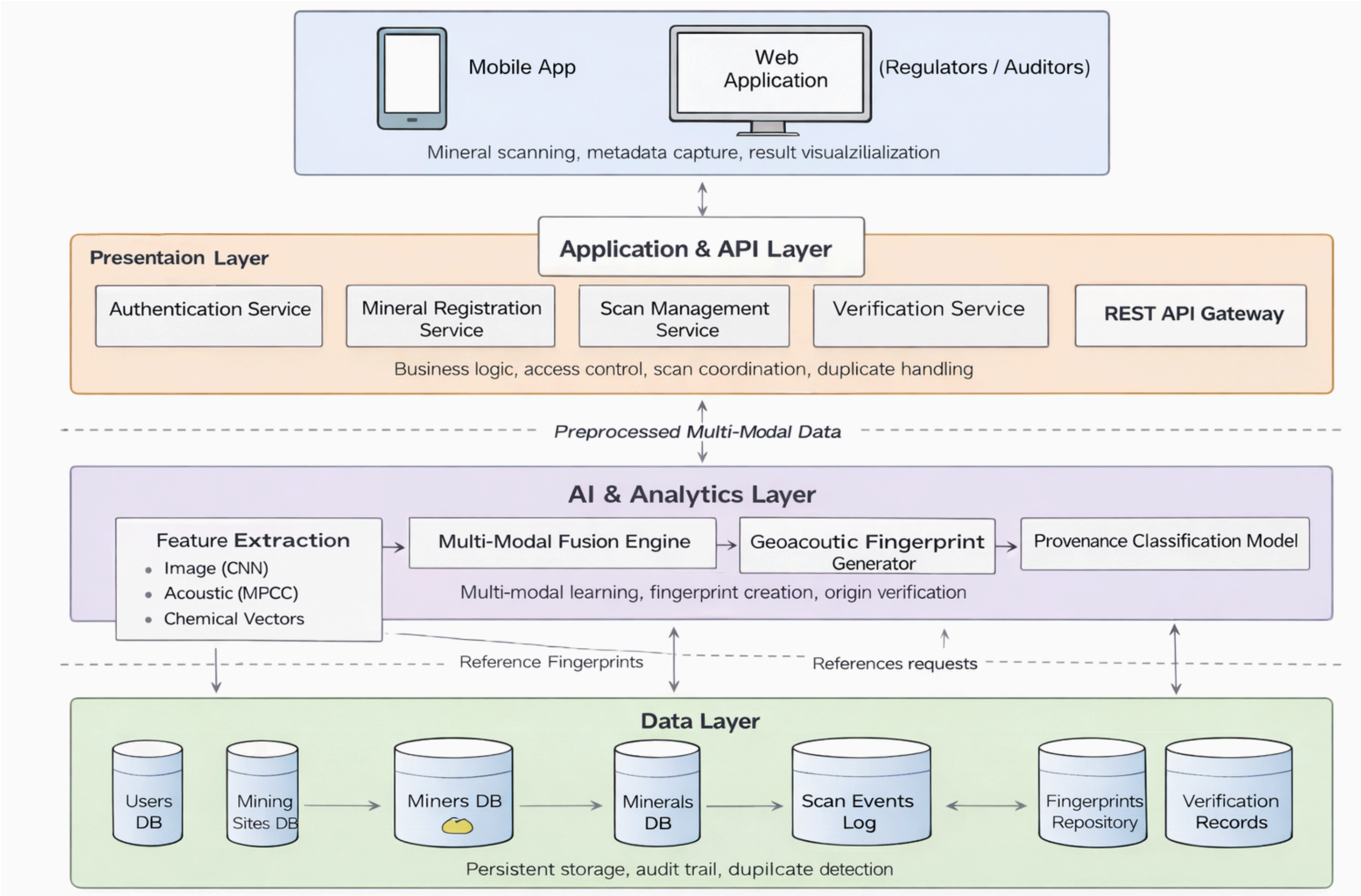
Performance	NFR-03	The system shall return verification results within an acceptable response time for field use.
Scalability	NFR-04	The system shall support an increasing number of minerals, users, and verification records without performance degradation.
Reliability	NFR-05	The system shall reliably store and retrieve verification and fingerprint data without data loss.
Maintainability	NFR-06	The system shall be modular to allow easy updates to machine learning models and system components.
Interoperability	NFR-07	The system shall support standard data formats and APIs for integration with external systems.
Data Integrity	NFR-08	The system shall prevent unauthorized modification of stored fingerprints and verification records
Availability	NFR-09	The system shall remain operational under low-bandwidth and intermittent connectivity conditions.
Portability	NFR-10	The system shall be deployable across web and mobile platforms without major redesign.

The non-functional requirements define the quality attributes and operational constraints of the system, ensuring security, usability, performance, scalability, reliability, and feasibility in low-resource environments**.**

3.4 System Architecture

The proposed system adopts a multi-layered architecture designed to support mineral registration, scanning, fingerprint generation, verification, and auditability. The architecture integrates user-facing applications, backend services, and a multi-modal AI engine to ensure secure, scalable, and scientifically grounded verification.

The system architecture consists of four main layers:



System Architecture of the AI-Powered Geoacoustic Fingerprinting System

Presentation Layer:
Provides web and mobile interfaces for inspectors, regulators, and auditors. This layer supports mineral scanning, metadata capture, and visualization of verification results.

Application and API Layer:
Handles business logic, authentication, mineral registration, scan management, verification coordination, and communication between the user interface and the AI engine.

AI and Analytics Layer:
Implements the core intelligence of the system, including feature extraction, multi-modal fusion, fingerprint generation, and provenance classification.

Data Layer:
Stores users, mining sites, registered minerals, fingerprints, scan events, and verification records. This layer supports traceability, auditability, and duplicate detection.

The layered architecture ensures separation of concerns, maintainability, and scalability while remaining feasible for deployment in low-infrastructure environments.

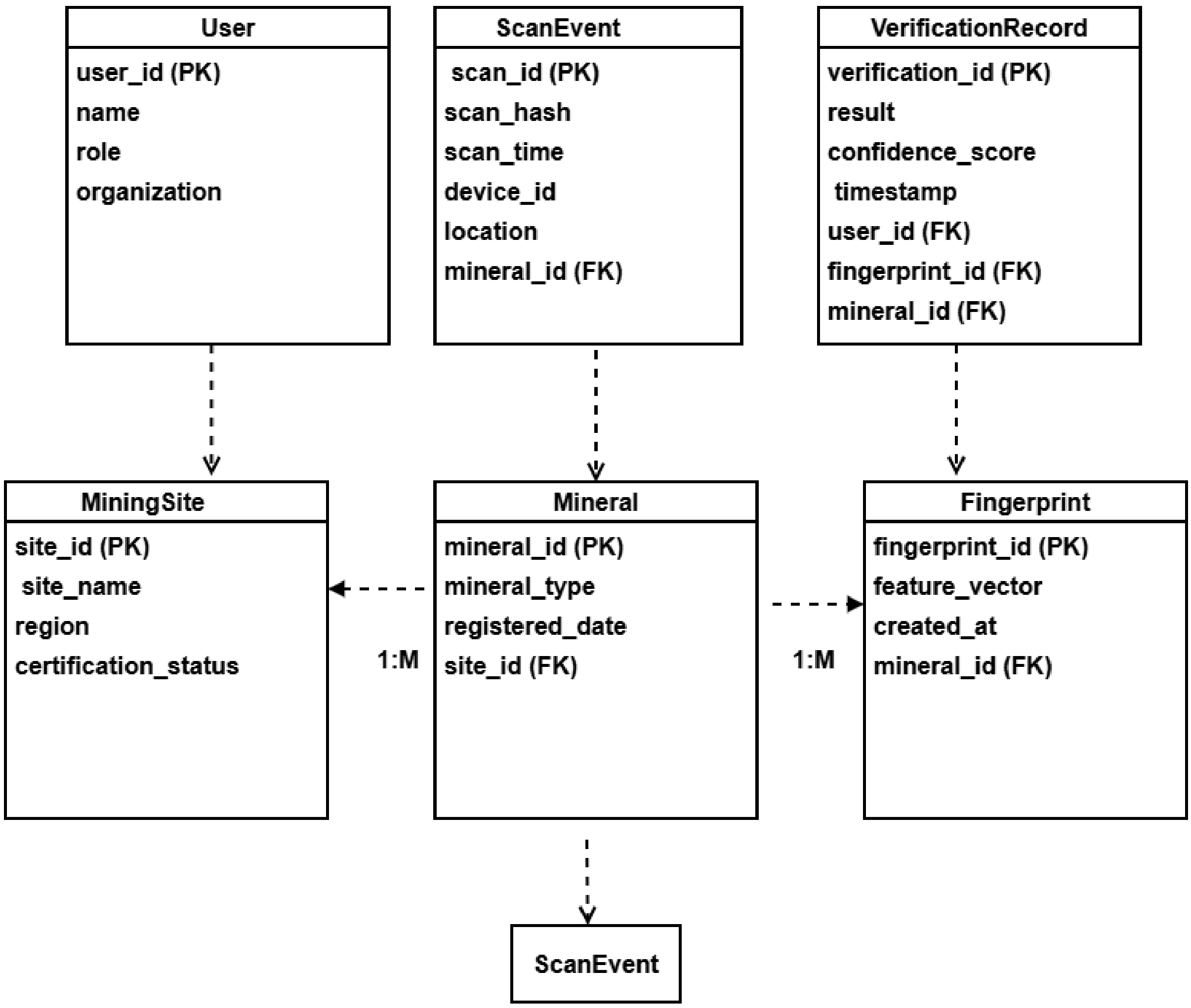
3.5 UML Diagrams

This section presents the Unified Modeling Language (UML) diagrams used to model the structural and behavioral aspects of the proposed AI-Powered Geoacoustic Fingerprinting

System. The UML diagrams provide a formal representation of system data structures, object-oriented design, user interactions, and process flows, ensuring clarity, consistency, and traceability between system requirements and implementation.

3.5.1 Entity Relationship Diagram (ERD)

The Entity Relationship Diagram (ERD) defines the core data entities and their relationships required to support mineral registration, verification, and auditability within the system. It focuses on how data is structured, stored, and linked to ensure integrity and traceability.



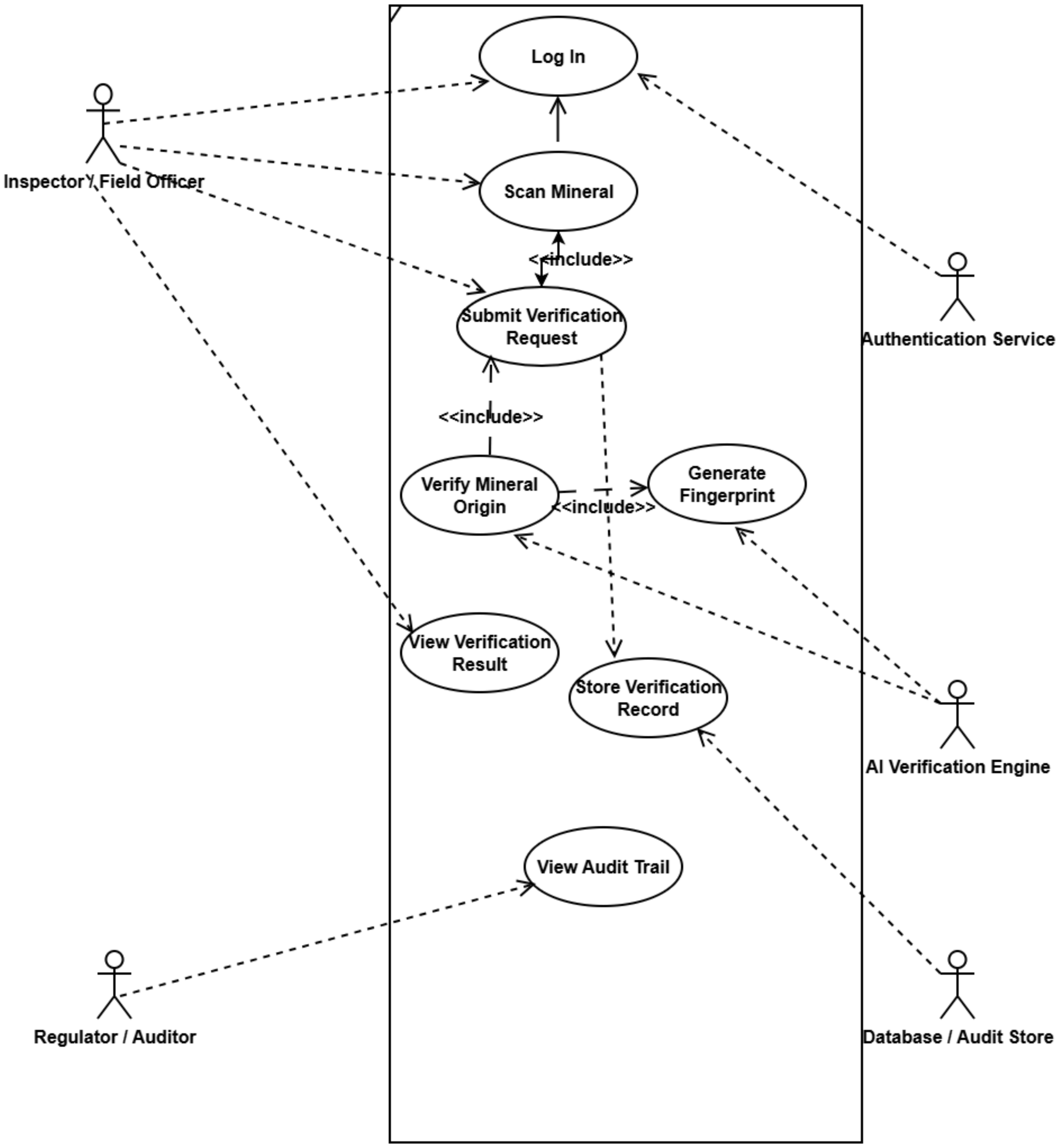
The key entities include MiningSite, Mineral, Fingerprint, ScanEvent, VerificationRecord, and User. A MiningSite registers one or more Minerals, establishing a clear link between mineral samples and certified extraction locations. Each Mineral may be associated with multiple Fingerprints, allowing the system to store geoacoustic fingerprints generated at different times or from different scans.

ScanEvent captures every instance in which a mineral is scanned, including contextual metadata such as time, location, and device information. These scan events support duplicate detection by ensuring that repeated scans of the same mineral can be identified. VerificationRecord stores the outcome of each verification process, linking together the User who initiated the verification, the Fingerprint used for comparison, and the associated Mineral. The User entity represents authorized system actors such as inspectors or regulators.

Collectively, these relationships enforce referential integrity, enable unique mineral identification, support duplicate detection, and provide a complete and auditable history of all verification activities within the system.

3.5.2 Use Case Diagram

The Use Case Diagram illustrates the functional interactions between external actors and the system, providing a high-level view of system behavior from the user's perspective. It focuses on what the system does rather than how it is implemented.



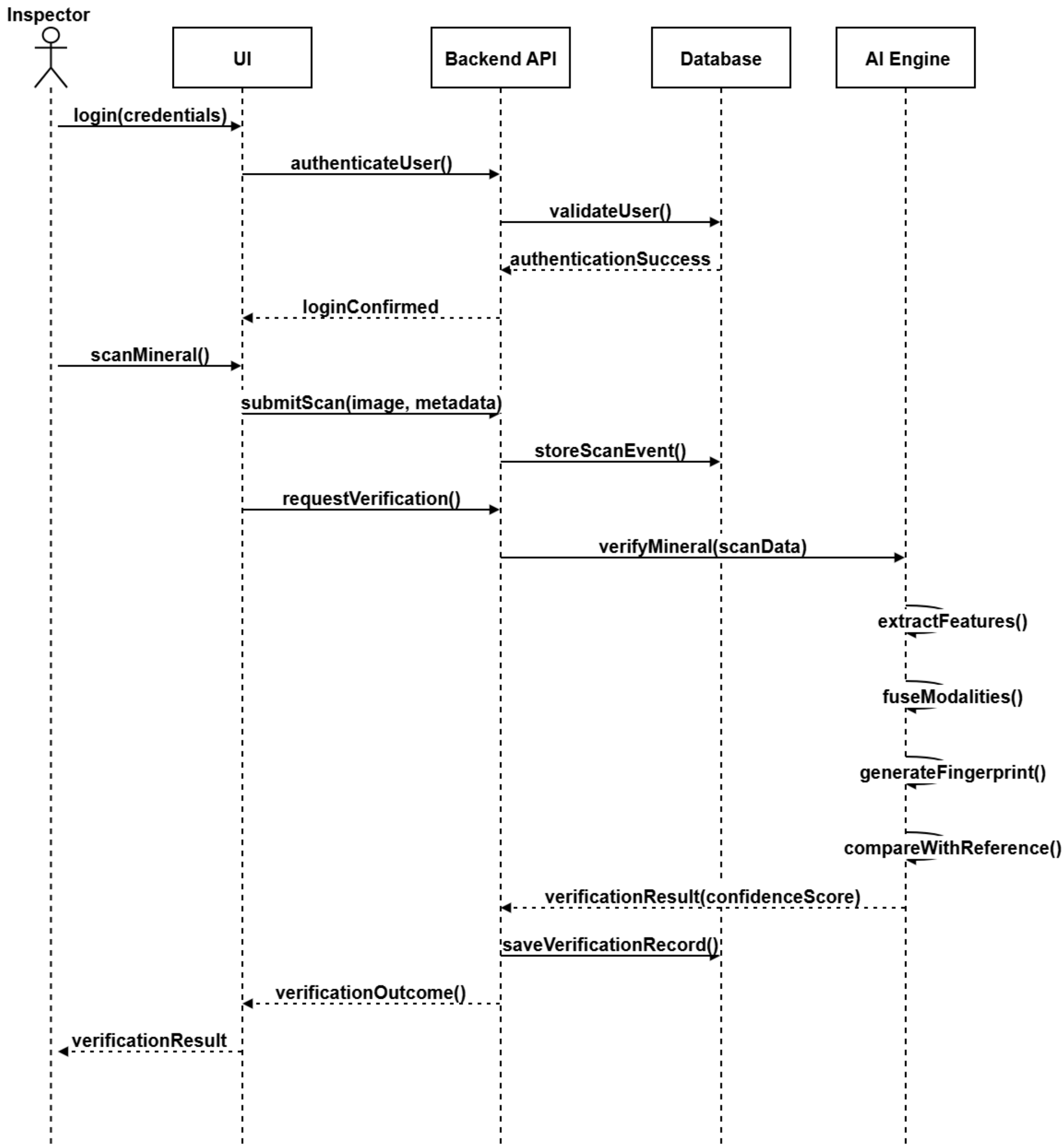
The primary actors include Inspector/Field Officer, Regulator/Auditor, and external services such as the Authentication Service and AI Verification Engine. The main use cases include Log In, Scan Mineral, Submit Verification Request, Verify Mineral Origin, Generate Fingerprint, View Verification Result, Store Verification Record, and View Audit Trail.

Authentication is a prerequisite for all verification-related actions, ensuring secure access. The Scan Mineral use case leads to Submit Verification Request, which in turn includes Verify Mineral Origin and Generate Fingerprint. Verification results are displayed to the user and stored for auditing purposes. Regulators and auditors can access the View Audit Trail use case to review historical verification records.

This diagram demonstrates how system functionality aligns with the defined functional requirements and clearly shows role-based access and system responsibilities.

3.5.3 Sequence Diagram

The Sequence Diagram models the dynamic behavior of the system by illustrating the chronological flow of interactions between actors and system components during a mineral verification process. It emphasizes message order, system coordination, and responsibility flow.



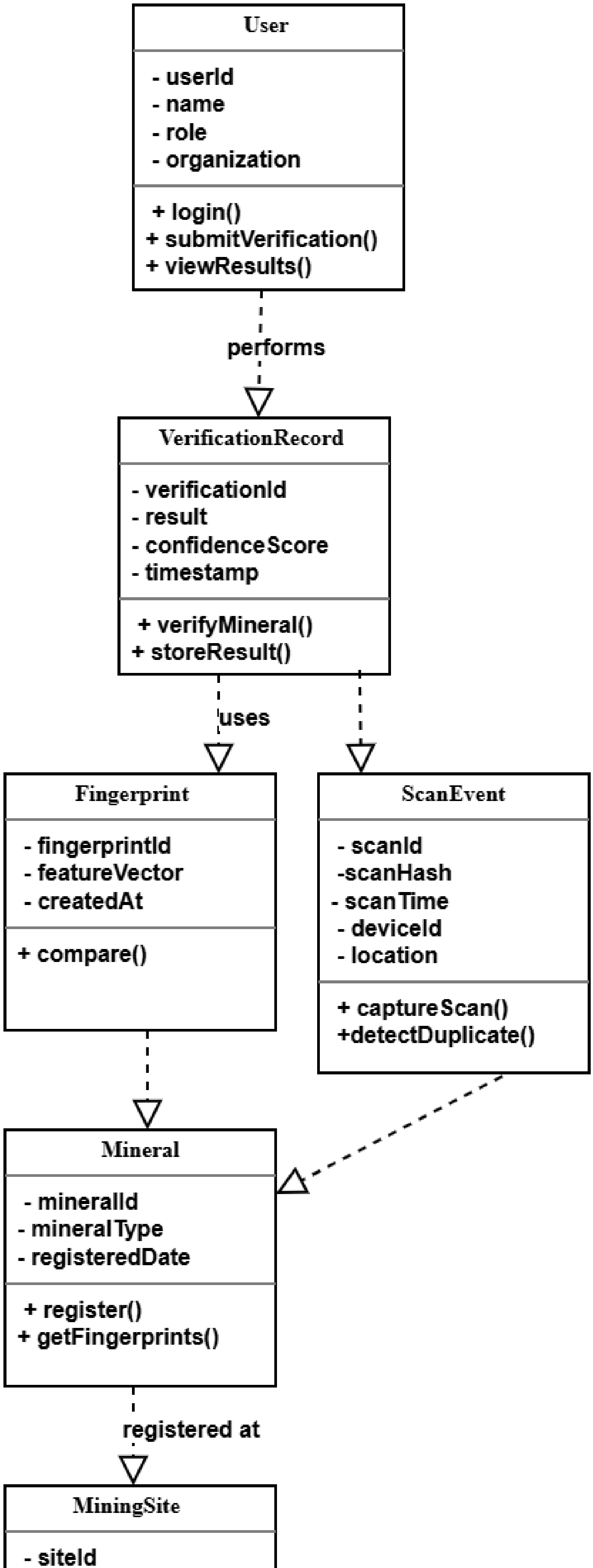
The sequence begins when a User logs into the system via the web or mobile interface. Authentication requests are forwarded to the backend API and validated against stored user credentials. Once authenticated, the user initiates a mineral scan, which generates a ScanEvent and sends scan data to the backend.

The backend invokes the AI Verification Engine, which performs feature extraction, multi-modal data fusion, fingerprint generation, and fingerprint comparison against registered references. The verification result and confidence score are returned to the backend, stored as a VerificationRecord, and sent back to the user interface for display.

This sequence highlights the end-to-end verification workflow, showing how user actions, backend services, AI components, and data storage work together to deliver reliable and auditable mineral provenance verification.

3.5.4 Class Diagram

The Class Diagram models the object-oriented structure of the system, illustrating how system components interact at the software level. It defines the attributes and operations of each class and shows how responsibilities are distributed across the system.


User
- userId
- name
- role
- organization
+ login()
+ submitVerification()
+ viewResults()
performs
VerificationRecord
- verificationId
- result
- confidenceScore
- timestamp
+ verifyMineral()
+ storeResult()
uses
Fingerprint
- fingerprintId
- featureVector
- createdAt
+ compare()
ScanEvent
- scanId
-scanHash
- scanTime
- deviceId
- location
+ captureScan()
+detectDuplicate()
Mineral
- mineralId
- mineralType
- registeredDate
+ register()
+ getFingerprints()
registered at
MiningSite
- siteId

Key classes include User, Mineral, MiningSite, Fingerprint, ScanEvent, VerificationRecord, and MLModel. The User class represents system actors who authenticate into the platform and initiate verification processes. The ScanEvent class encapsulates the scanning operation, capturing metadata and generating scan hashes to support duplicate detection.

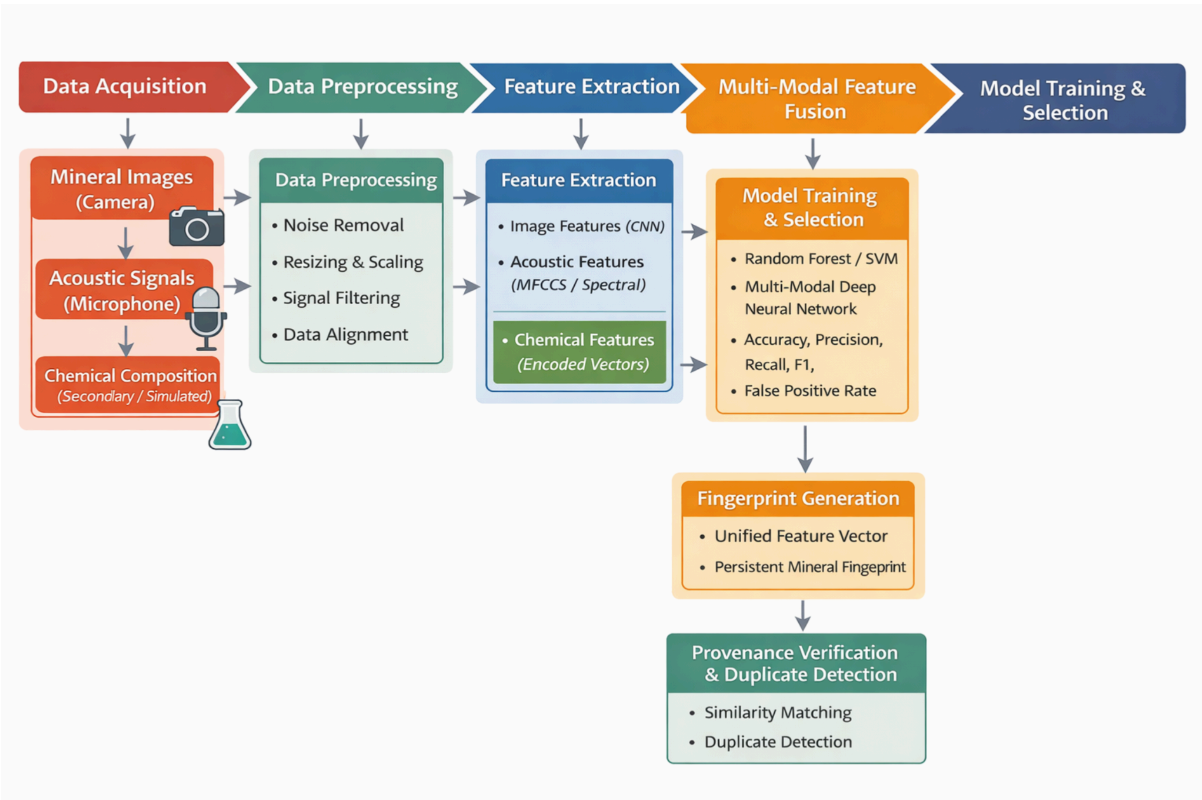
The Fingerprint class stores fused multi-modal feature vectors generated from mineral scans, while the Mineral class acts as the central domain object linking scans, fingerprints, and verification records. Each Mineral is registered at exactly one MiningSite, ensuring geographical traceability. The VerificationRecord class records verification results, confidence scores, and timestamps, forming an auditable decision log.

The MLModel class represents the artificial intelligence component of the system. It is responsible for feature extraction, multi-modal fusion, fingerprint generation, and provenance classification. This separation of machine learning logic from domain and service classes promotes modularity, scalability, and maintainability.

Overall, the class structure clearly separates concerns between user interaction, data management, and AI processing, supporting a robust and extensible system design.

3.6 Machine Learning Pipeline Architecture

This section describes the end-to-end machine learning (ML) pipeline used in the proposed AI-Powered Geoacoustic Fingerprinting System. The pipeline explains how raw mineral data is acquired, processed, transformed into features, modeled using multiple machine learning techniques, and finally used to generate and verify unique mineral fingerprints. The design prioritizes robustness, interpretability, and feasibility within low-resource and conflict-affected environments.



3.6.1 Data Acquisition

The system relies on multi-modal data sources to capture intrinsic mineral characteristics. Data acquisition is designed to be software-driven and minimally dependent on specialized hardware.

The primary data sources include:

- Visual Data:
  High-resolution images of mineral samples captured using mobile or web device cameras. These images capture surface texture, color variation, grain structure, and visible inclusions.
- Acoustic Data:
  Acoustic signals generated by tapping or stimulating mineral samples. The recorded signals reflect resonance, density, and internal structure characteristics unique to each mineral.
- Chemical Composition Data:
  Secondary and simulated geochemical datasets representing elemental composition and trace element profiles. These datasets are sourced from publicly available geological databases and peer-reviewed literature.

All data inputs are tagged with metadata such as scan time, device ID, location, and associated mining site.

3.6.2 Data Preprocessing

Before model training and inference, raw data undergoes preprocessing to ensure consistency, noise reduction, and compatibility across modalities.

- Image Preprocessing:
  Image resizing, normalization, noise reduction, and color-space standardization are applied. Data augmentation techniques such as rotation and scaling are used to improve model generalization.
- Acoustic Signal Preprocessing:
  Acoustic signals are filtered to remove background noise and normalized to a standard sampling rate. Time-domain signals are transformed into frequency-domain representations.
- Chemical Data Preprocessing:
  Chemical vectors are normalized, missing values handled, and irrelevant or redundant attributes removed to reduce dimensionality.

Preprocessed data is then synchronized across modalities to ensure each mineral sample is represented consistently.

3.6.3 Feature Extraction

Feature extraction converts raw data into meaningful numerical representations.

- Visual Features:
  Convolutional Neural Networks (CNNs) are used to extract high-level visual features such as texture patterns and spatial structures.

- Acoustic Features:
  Signal processing techniques such as Mel-Frequency Cepstral Coefficients (MFCCs), spectral centroid, and resonance features are extracted from acoustic data.

- Chemical Features:
  Elemental concentration vectors and statistical descriptors are used to represent chemical composition.

Each modality produces an independent feature vector that captures complementary aspects of mineral identity.

3.6.4 Multi-Modal Feature Fusion

To create a unified representation, extracted features from all modalities are combined using a feature fusion strategy. Intermediate-level fusion is employed, where modality-specific features are first learned independently and then concatenated into a single composite vector.

This approach allows the system to learn cross-modal correlations while remaining resilient to partial or noisy data from any single modality.

3.6.5 Model Experimentation and Selection

Multiple machine learning models are experimented with to identify the most effective approach for mineral fingerprinting and provenance verification.

The models evaluated include:

- Traditional Machine Learning Models:
  - Support Vector Machines (SVM)
  - Random Forest Classifiers
  - Gradient Boosting Models
- Deep Learning Models:
  - Convolutional Neural Networks (CNNs) for image features
  - Multi-layer Perceptrons (MLPs) for fused feature vectors

- Hybrid CNN–MLP architectures for multi-modal learning

Each model is trained using labeled reference data representing certified mining sites and mineral samples.

3.6.6 Model Evaluation

Models are evaluated using quantitative performance metrics to ensure reliability and scientific rigor.

The evaluation metrics include:

- Accuracy
- Precision
- Recall
- F1-score
- False Positive Rate

Cross-validation techniques are applied to assess generalization performance and reduce overfitting. The model that achieves the best balance between accuracy, robustness, and computational efficiency is selected as the final verification model.

3.6.7 Fingerprint Generation and Verification

The selected model generates a geoacoustic fingerprint, represented as a high-dimensional feature vector uniquely associated with each mineral sample. These fingerprints are stored in the system database as reference identities.

During verification, newly generated fingerprints are compared against stored references using similarity measures. If similarity exceeds a defined threshold, the mineral origin is verified; otherwise, it is flagged as unregistered or potentially fraudulent. Scan hashes and fingerprint similarity jointly support duplicate detection.

3.6.8 Integration with System Architecture

The machine learning pipeline is fully integrated into the AI and Analytics Layer of the system architecture. Feature extraction, fusion, model inference, and fingerprint comparison are exposed through backend services and invoked during each verification request.

This integration ensures that AI-driven verification operates seamlessly within the overall system workflow while maintaining modularity and scalability.

3.6.9 Summary of the Machine Learning Pipeline

In summary, the machine learning pipeline transforms raw multi-modal mineral data into scientifically grounded geoacoustic fingerprints through systematic preprocessing, feature extraction, data fusion, model experimentation, and evaluation. This pipeline enables intrinsic, data-driven mineral provenance verification that complements existing traceability mechanisms and is suitable for deployment in low-resource environments.

3.7 Development Tools

The system is developed using open-source tools to ensure accessibility and reproducibility:

- Programming Language: Python
- Machine Learning Frameworks: TensorFlow, PyTorch
- Image Processing: OpenCV
- Signal Processing: Librosa, SciPy
- Backend Framework: FastAPI
- Frontend Technologies: HTML, CSS, JavaScript, Flutter (Dart)
- Database: SQLite or PostgreSQL
- Version Control: Git

These tools support rapid prototyping, efficient experimentation, and reliable system development within an academic setting.

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